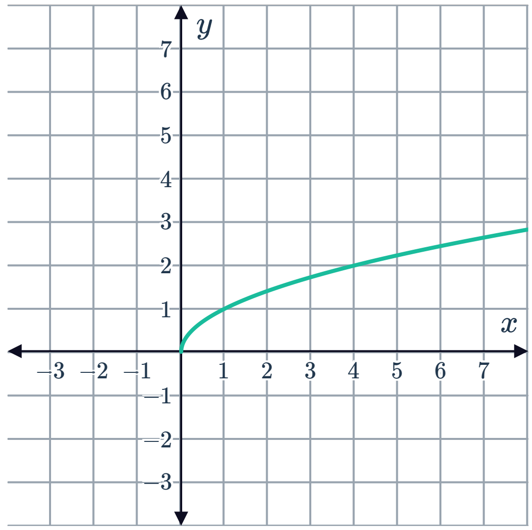
Graphs of square root functions

1

a No

b ∞

c



d As x increases, the function increases at a slower and slower rate.

e $(0,0)$

f No

g No

h $y=5\sqrt{x}$ increases more rapidly than $y=\sqrt{x}$.

2

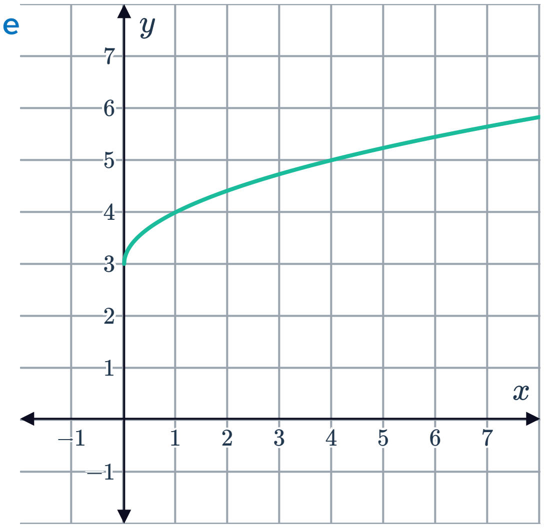
a No

b The function decreases at a slower and slower rate.

3

a No   **b** ∞ **c** $y=3$ **d** 0

e



4

a $x \leq 0$

b $y \geq 0$

c ∞

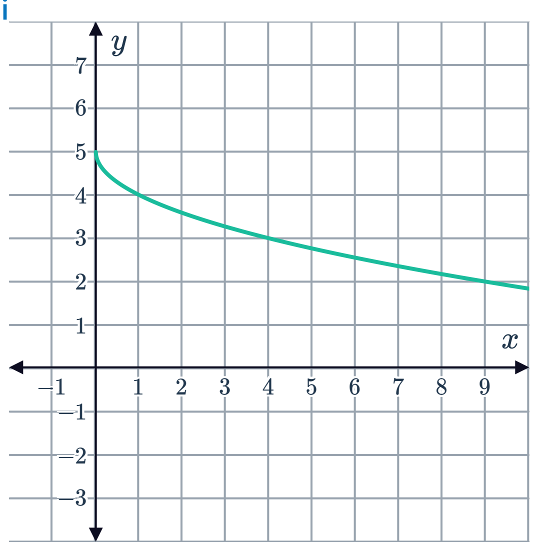
5

a

i $x \geq 0$

ii $y \leq 5$

iii

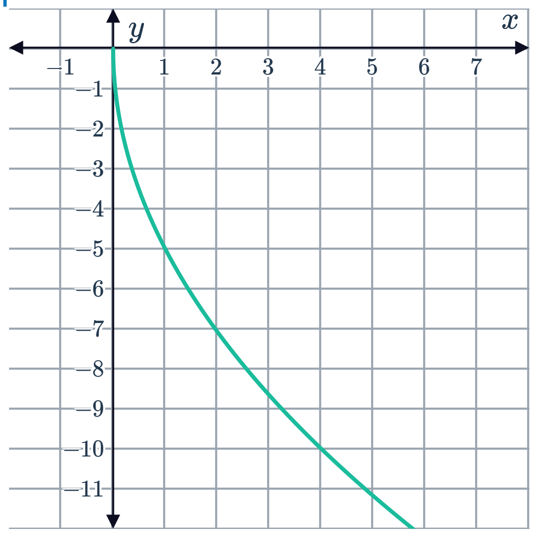


b

i $x \geq 0$

ii $y \leq 0$

iii

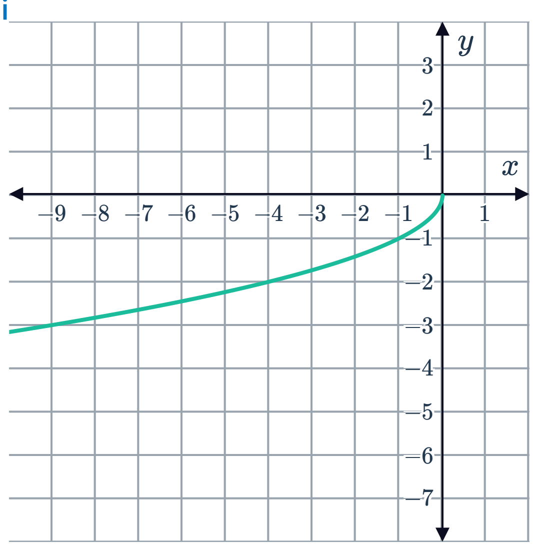


c

i $x \leq 0$

ii $y \leq 0$

iii



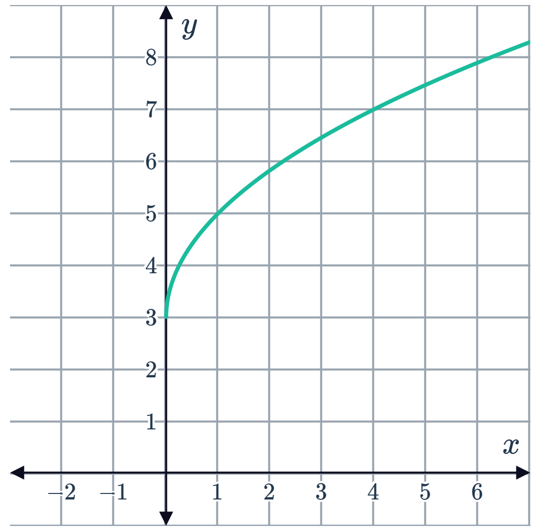
6

a Increasing

b More steep

c $(0, 3)$

d



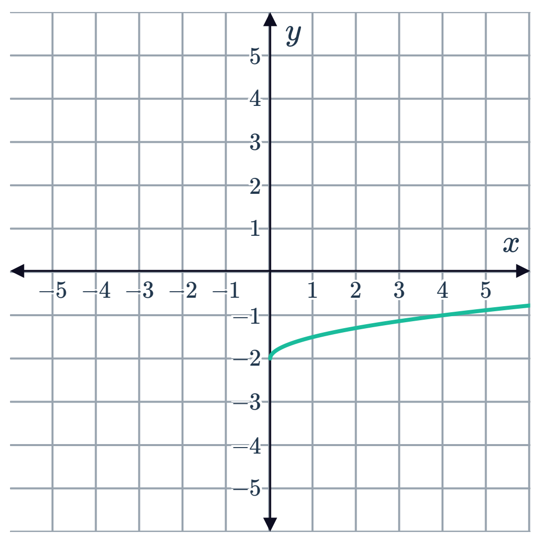
7

a Increasing

b Less steep

c $(0, -2)$

d



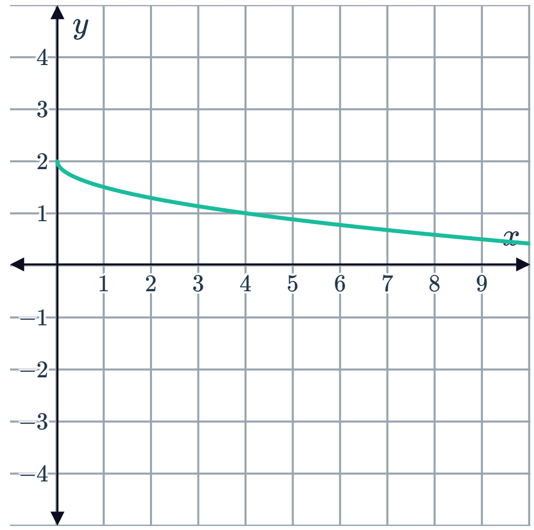
8

a Decreasing

b Less steep

c $(0, 2)$

d



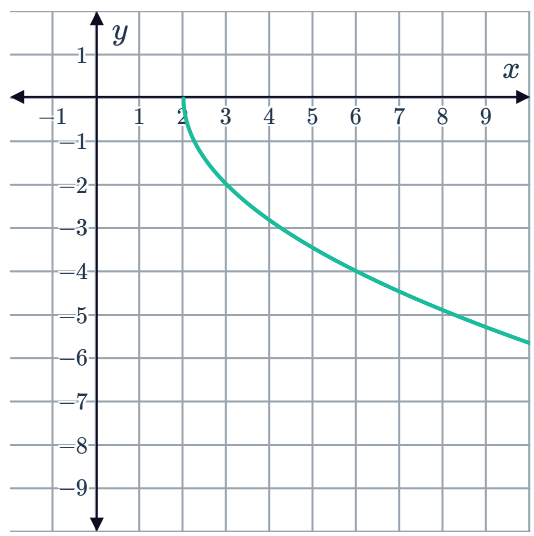
9

a Decreasing

b More steep

c $(2, 0)$

d



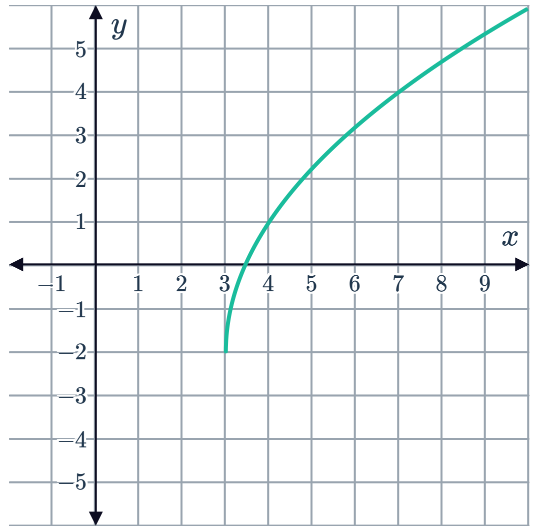
10

a Increasing

b More steep

c $(3, 2)$

d



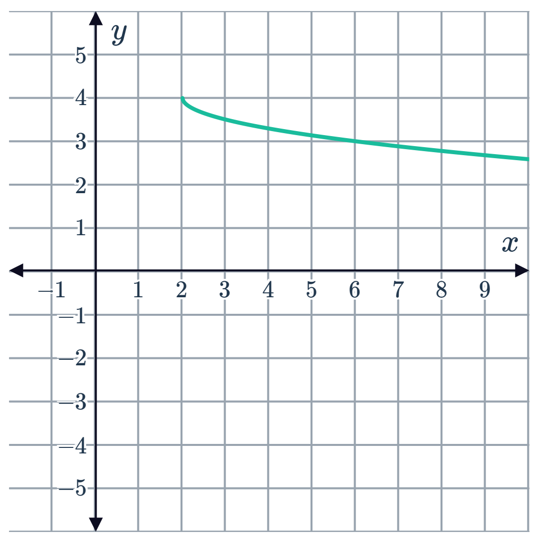
11

a Decreasing

b Less steep

c $(2, 4)$

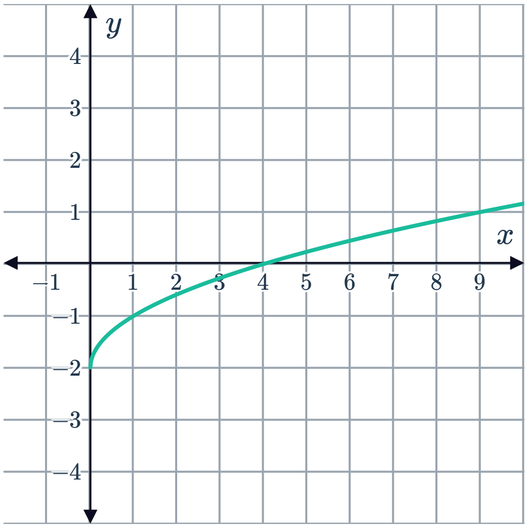
d



12

a

i

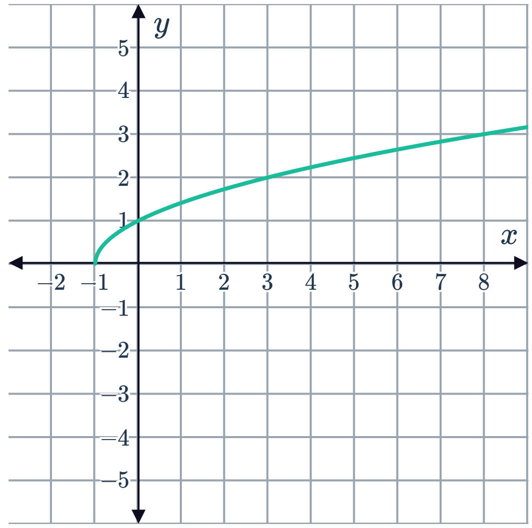


ii $x \geq 0$

iii $y \geq -2$

b

i

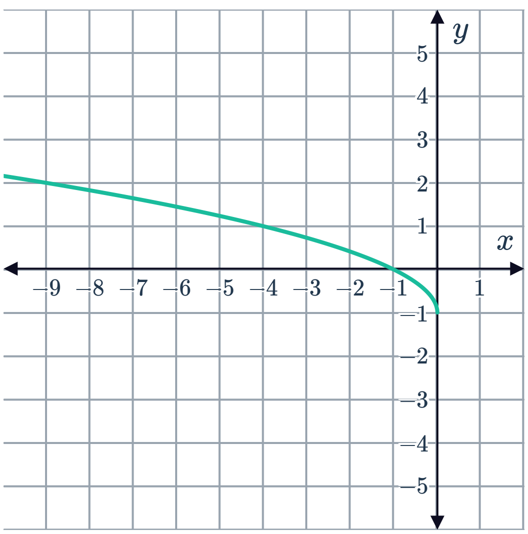


ii $x \geq -1$

iii $y \geq 0$

c

i

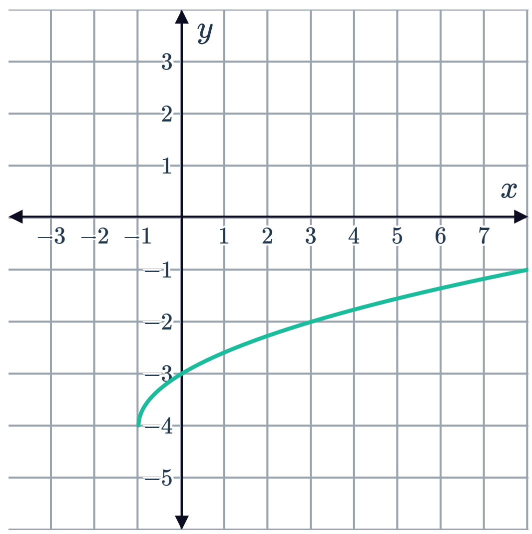


ii $x \leq 0$

iii $f(x) \geq -1$

d

i

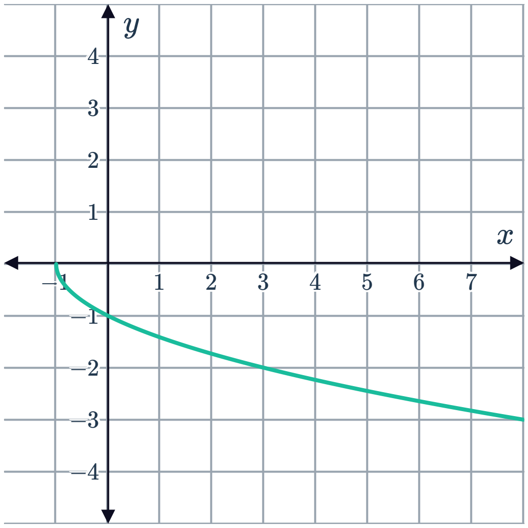


ii $x \geq -1$

iii $y \geq -4$

e

i

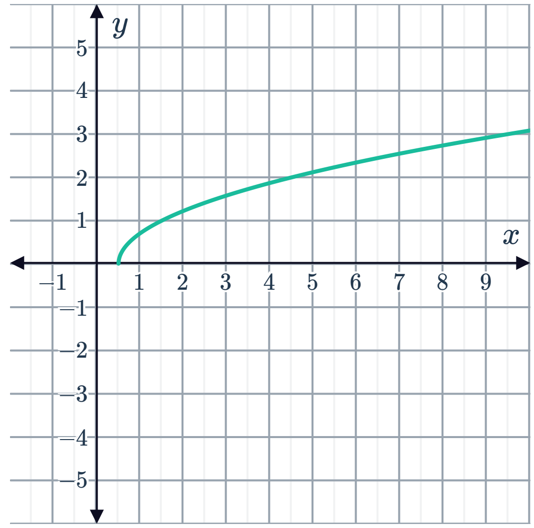


ii $x \geq -1$

iii $y \leq 0$

f

i

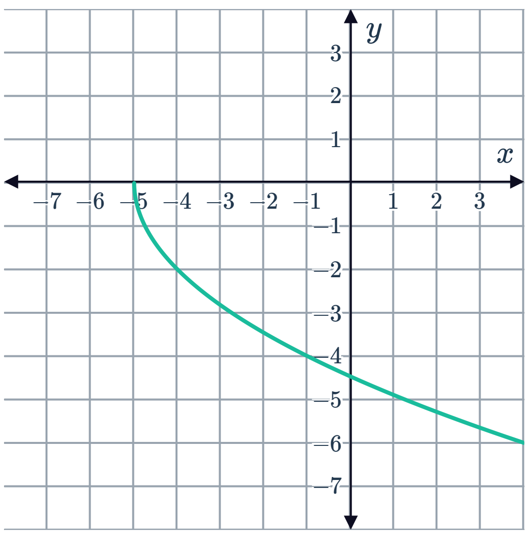


ii $x \geq 2$

iii $y \geq 0$

g

i

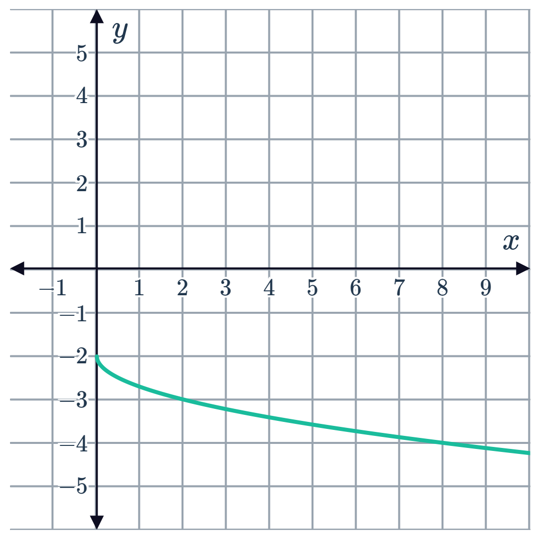


ii $x \geq -5$

iii $y \leq 0$

h

i

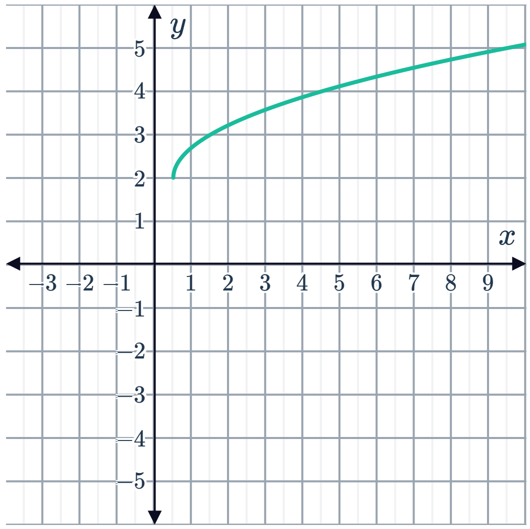


ii $x \geq 0$

iii $y \leq -2$

i

i



ii $x \geq 1$

iii $y \geq 2$

Characteristics of square root functions

13

a No

b No

c No

d No

e Yes

14 Domain: $x \geq 0$, range: $y \geq -2$

15

a $x \geq 5$

b $y \geq 0$

c

i Yes

ii No

iii No

16

a 6

b $x \leq 0$

c $y \geq 6$

17 $[2, \infty)$

Transformations

18

a $y = 2$

b Vertical shift 2 units up

c $f(x) = \sqrt{x} + 2$

19

a $x = -2$

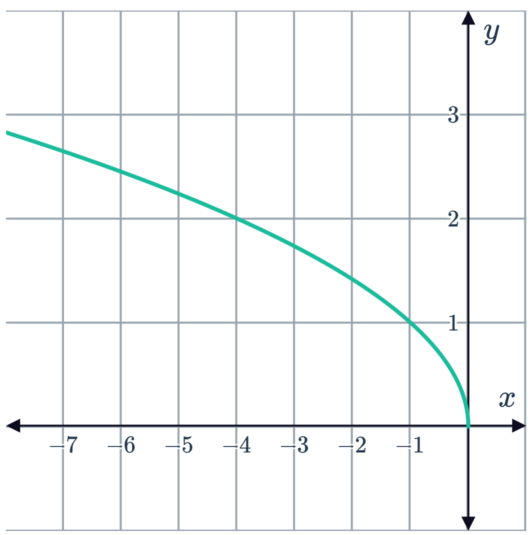
b Horizontal shift 2 units to the left

c $f(x) = \sqrt{x+2}$

20

a

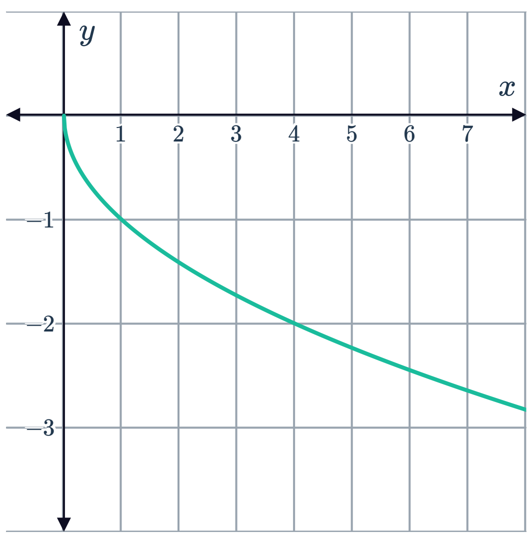
i



ii $y = \sqrt{-x}$

b

i



ii $y = -\sqrt{x}$

21

a $x \leq 0$

b $y = 3\sqrt{-x}$

22

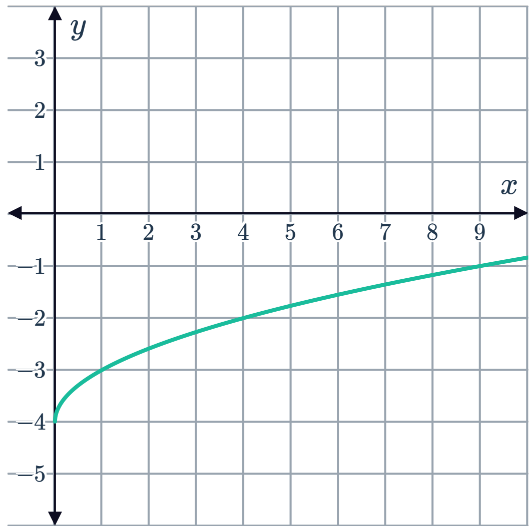
a 3

b Downwards

23

a The original function was translated down 4 units.

b



24 y increases more rapidly on $y = 3\sqrt{x}$ than on $y = \sqrt{x}$

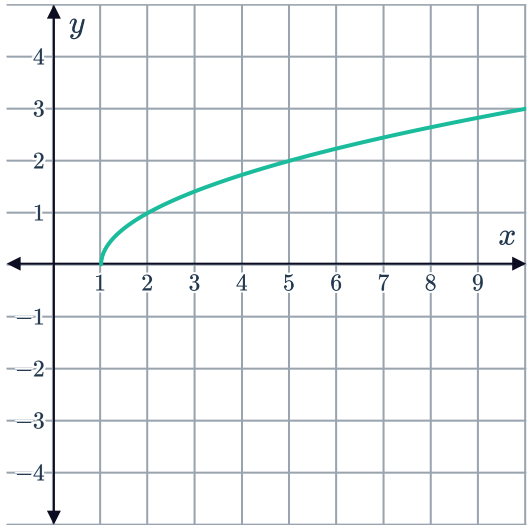
25

a $(0, -2)$

b $(0, 3)$

26

a



b Horizontal translation 1 unit(s) to the right

27

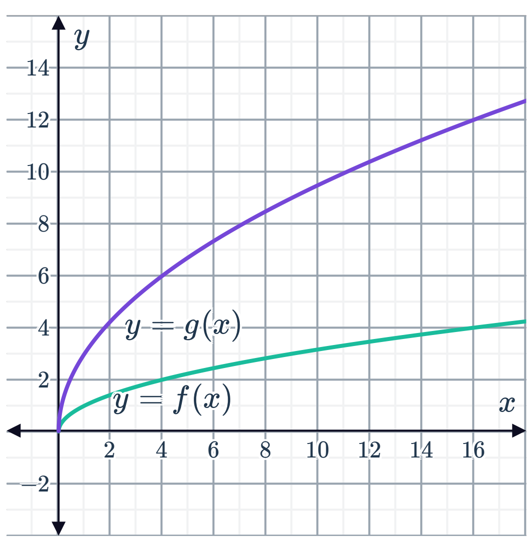
- Reflection across x-axis, then a translation 5 units down
- Translation 5 units up, then a reflection across x-axis

28

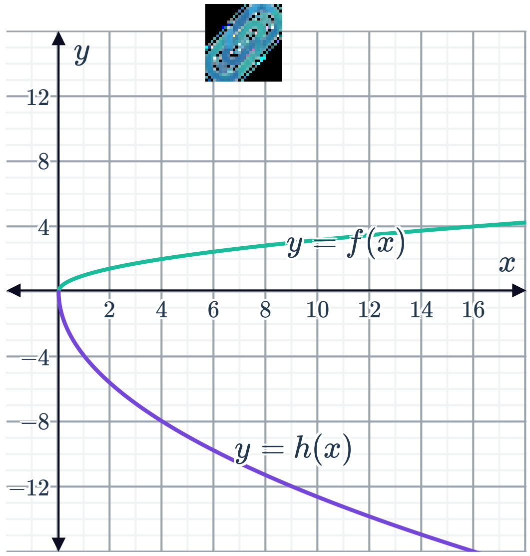
a

x	0	1	4	9	16
$y = f(x)$	0	1	2	3	4
$g(x) = 3f(x)$	0	3	6	9	12
$h(x) = -4f(x)$	0	-4	-8	-12	-16

b



d

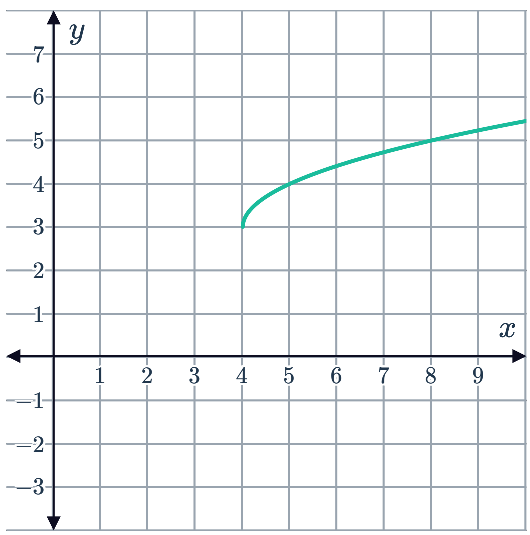


e Vertically stretched by a factor of 4 and reflected across the x-axis

29

a Translate the graph 4 units to the right and 3 units up.

b

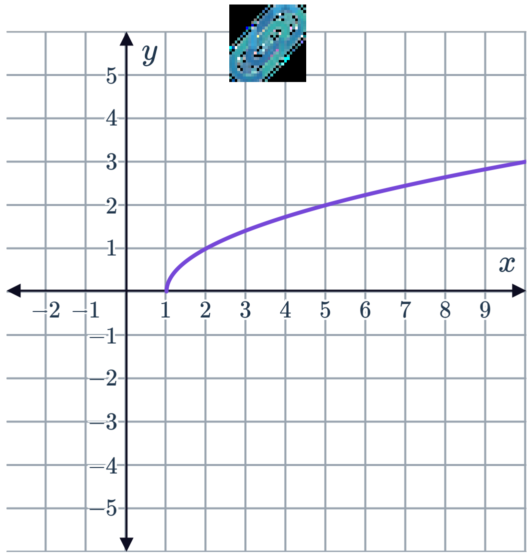


30

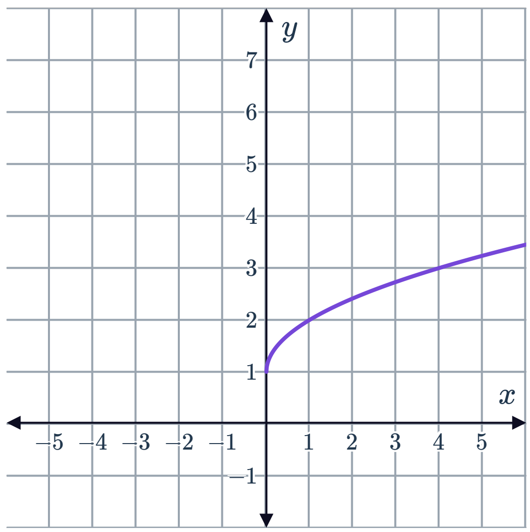
a $y = \sqrt{x - 5}$ **b** $y = -\sqrt{x} + 6$ **c** $y = -\frac{1}{2}\sqrt{x}$

31 $\frac{\sqrt{2x}}{4} + 0$

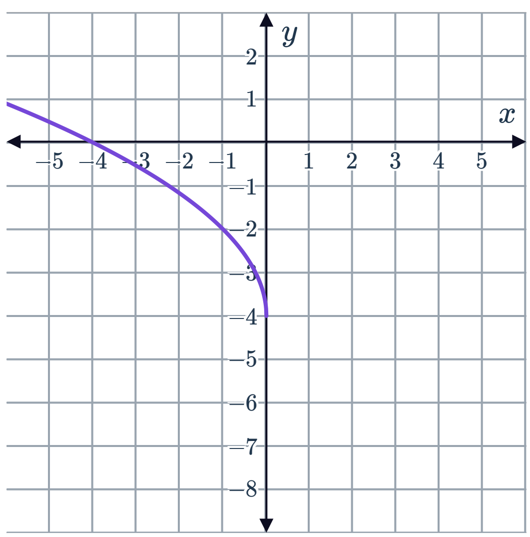
32


y
x
5
4
3
2
1
−1
−2
−3
−4
−5
−2 −1 1 2 3 4 5 6 7 8 9

33


y
x
7
6
5
4
3
2
1
−1
−5 −4 −3 −2 −1 1 2 3 4 5

34


y
x
2
1
−1
−2
−3
−4
−5
−6
−7
−8
−5 −4 −3 −2 −1 1 2 3 4 5

Applications

35

a $4\sqrt{5}$ miles

b 8.9 miles

36 1.57 seconds

37

a $12\sqrt{2}$ feet per second

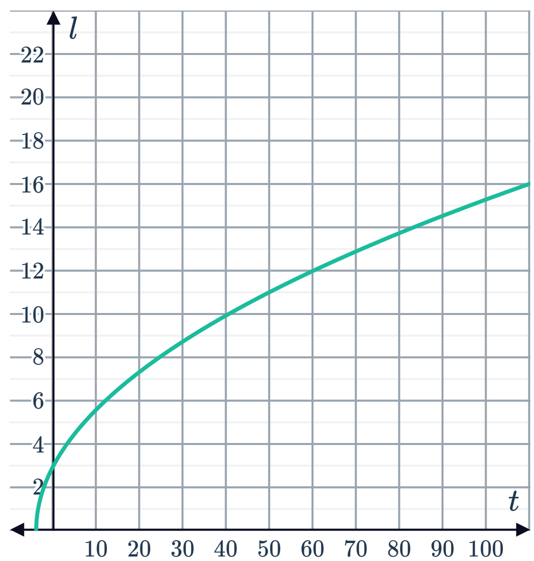
b 17 feet per second

38

a

Months (t)	0	60	96
Length (l)	3	12	15

b



c $t \geq 0$

d No

e 14.1 meters

f 0.7

39 $A = 14657415$

40

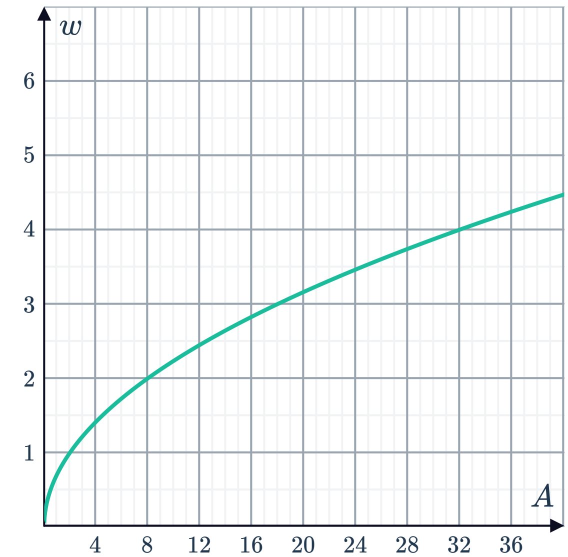
a $A = 2w^2$

b $w = \sqrt{\dfrac{A}{2}}$

c

Area (A)	0	2	8	18	32
Width (w)	0	1	2	3	4

d



e 50 units^2

f Above

41

a $D = -\sqrt{10}\ \text{t}$

b $-5\sqrt{2}$ meters

c $\sqrt{2}$ m/s

d $2 - \sqrt{2}$ m/s